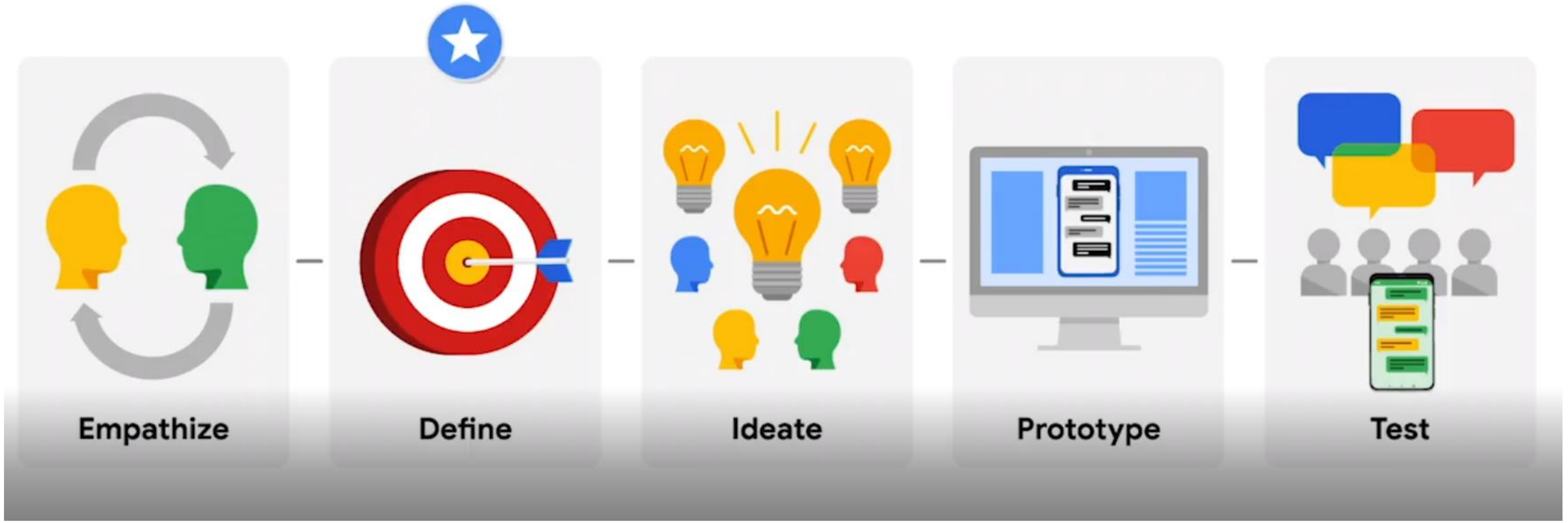
# Stage 2: Define—*State Your Users' Needs and Problems*

# 2. Define: problem statement

Once you've empathized with your users, you can move on to the second stage of the Design Thinking process and define the problem your users need you to solve.

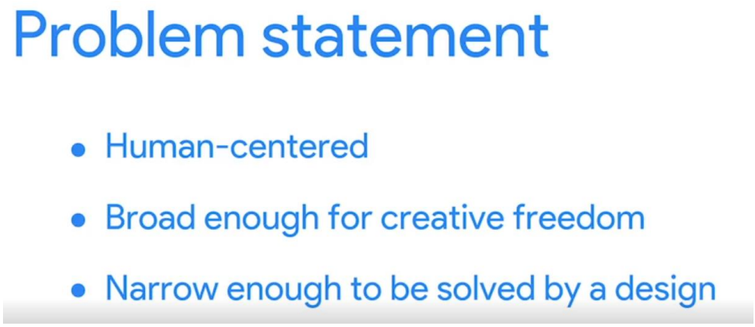


# 2. Define: problem statement

2. What is a problem statement?

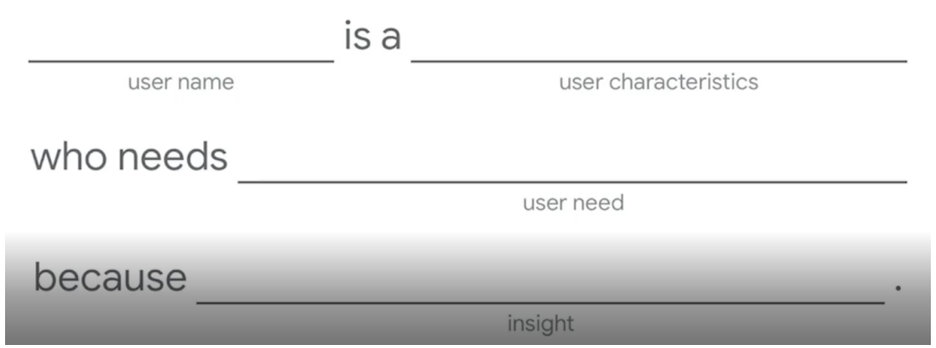
- A problem statement identifies the gap between the current state (i.e. the problem) and the desired state (i.e. the goal) of a process or product. Within the design context, you can think of the user problem as an unmet need. By designing a solution that meets this need, you can satisfy the user and ensure a pleasant user experience.
- A problem statement is a clear description of the user's needs that should be addressed. Problem statements align the team on which user problem to focus on, giving everyone a clear goal. A strong problem statement is human-centered. It should be broad enough to allow for some creative freedom, but narrow enough that it can actually be solved by a design solution.



# **Create** problem statement

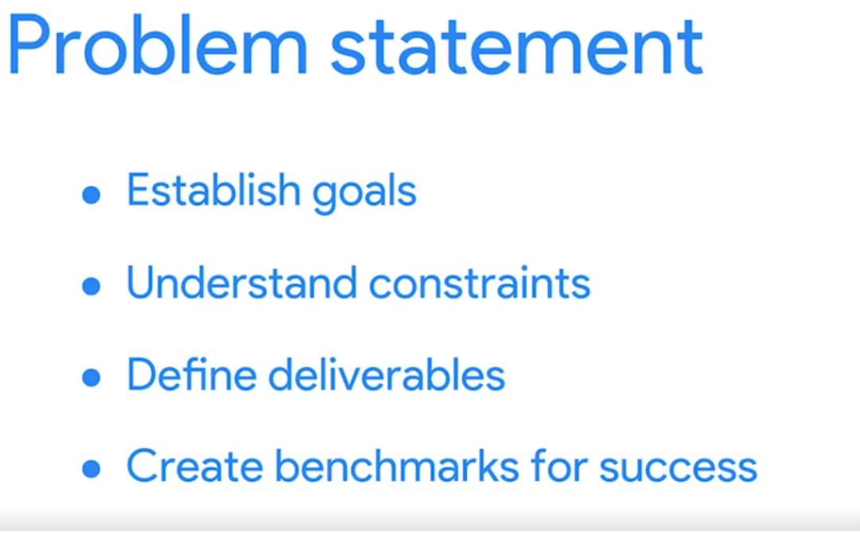
- Problem statements can be written using a simple formula. Start with the name of the user. Add a short description of the user's characteristics. Clearly describe the user's need and explain why the user has that need; in other words, an insight about the user.
- For example, Amal is the name of the user. His user characteristic is that he's an athlete. His user need is that he wants to sign up for a workout class. And the insight, or why he has the need, is that the workout classes are filling up quickly.






# **Create** problem statement

**1.Goal Establishment**: Problem statements set clear goals based on the user's real needs, ensuring that everyone on the design team understands and focuses on the same objectives.
**2.Understanding Constraints**: They highlight the obstacles preventing users from meeting their needs, providing a clearer path for addressing these challenges.
**3.Defining Deliverables**: Problem statements clarify the outcomes expected from solving the problem, making it clear what the solution should achieve.
**4.Creating Benchmarks for Success**: They establish criteria for success, enabling the team to measure when they have effectively solved the problem.




4.For instance, knowing you've succeeded in opening a door when you can see what's behind it simplifies and clarifies the goal, even for more complex design challenges.
Overall, an effective problem statement guides the design process, focusing on user-centric solutions and measurable outcomes.

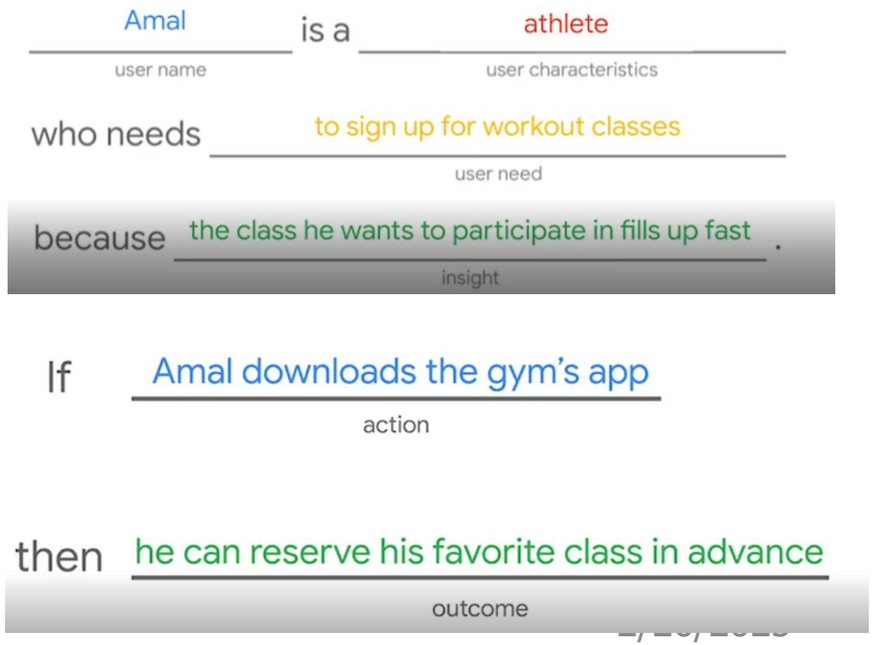
# Hypothesis statements

- Brainstorming solutions resembles formulating a hypothesis or an educated guess about solving a problem. In UX design, we articulate potential solutions through hypothesis statements.
- A hypothesis statement articulates our best educated guess about a potential solution to a design problem

## Hypothesis statement

Our best educated guess on what we think the solution to a design problem might be

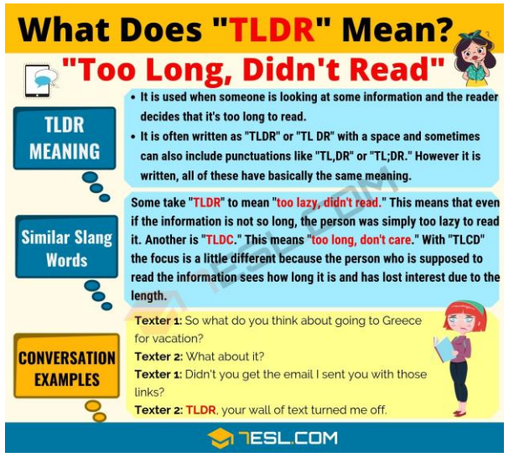
Amal needs an app that allows him to reserve his favorite classes in advance and notifies him of the first opportunity to sign up.



# Human factors

- The human factor describes the range of variables humans bring to their product interactions.
- For an example of design that considers these factors, all you need to do is check your email. The business email shorthand TL;DR has really caught on in the last couple of years. It's an acronym you might find at the start of a very long email. A TL;DR is a short, succinct summary that only gives you the email highlights you really need to know, without any excess content. So, what does TL;DR stand for? Too long; didn't read.




The email writer factors in the human tendencies of impatience, limited concentration, need for motivation, and limited memory. Not bad for four little letters.

# Human factors

- Here are psychological concepts that can help you design with the human factor in mind. Mental models are internal maps that allow humans to predict how something will work. Wher you face a door, your mental model tells you that the door can be opened.
- we'll go through some of the psychological principles that act on the user's subconscious as they interact with the product.

## Common human factors that inform design

- Impatience
- Limited memory
- Needing analogies
- Limited concentration
- Changes in need
- Needing motivation
- Prejudices
- Fears
- Making errors
- Misjudgment

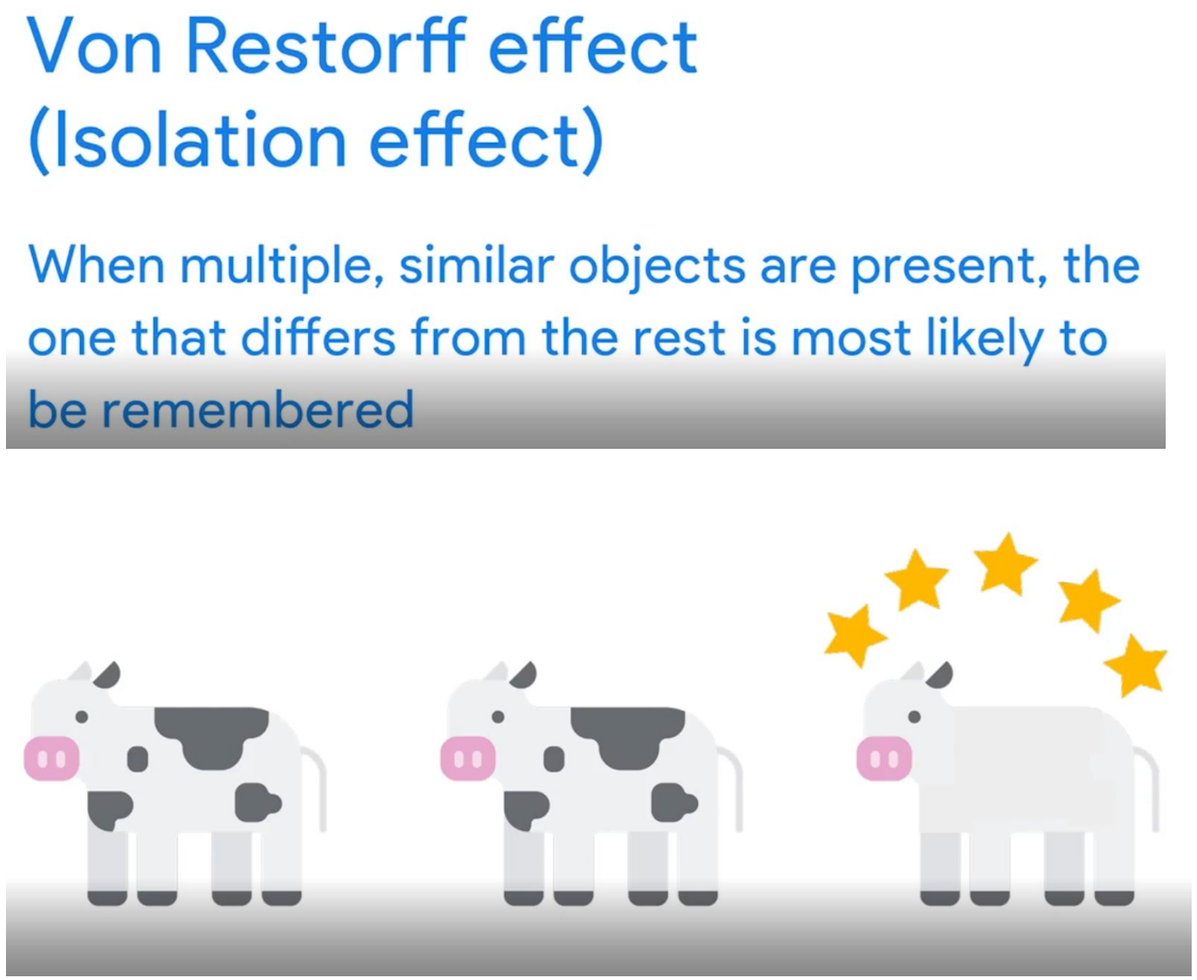
# Psychological phenomena

Additionally, the session explored several psychological phenomena relevant to UX design:

**1.Von Restorff Effect (Isolation Effect)**

- This phenomenon explains why items that stand out in a group are more likely to be remembered.

An example given was from children's spot-the-difference games, where a distinct item among similar ones catches attention and remains memorable.

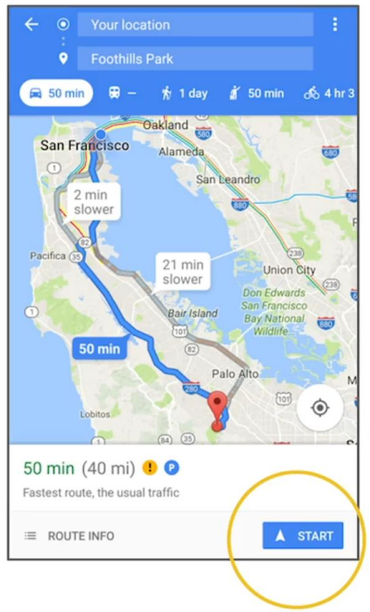


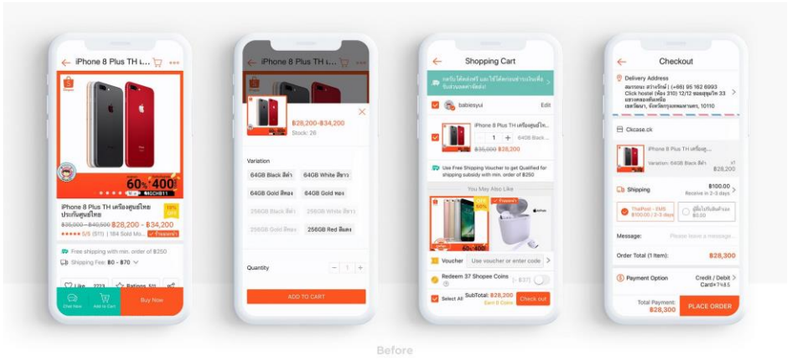
# Psychological phenomena

- In UX design, this is why the call-to-action buttons look different from the rest of the buttons on a site or app; because we want them to stand out.
- In case you're not familiar, a call-to-action, or CTA, is a visual prompt that tells the user to take action. For example, the Start button on Google Maps is bright blue on a white background, which makes it stand out. The second psychological phenomena, serial position effect, says that when people are given a list of items, they are more likely to remember the first few and the last few, while the items in the middle tend to blur. This is why most applications and websites position the most important user actions toward the far right or far left of a top navigation bar.

Call-to-action (CTA)

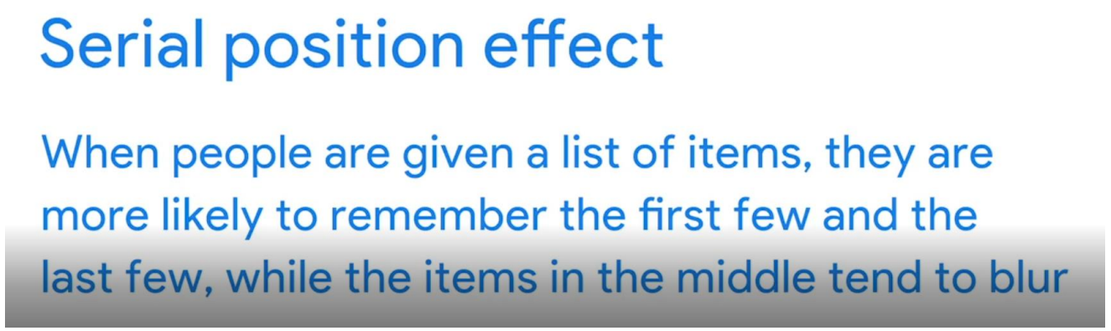
A visual prompt that tells the user to take action

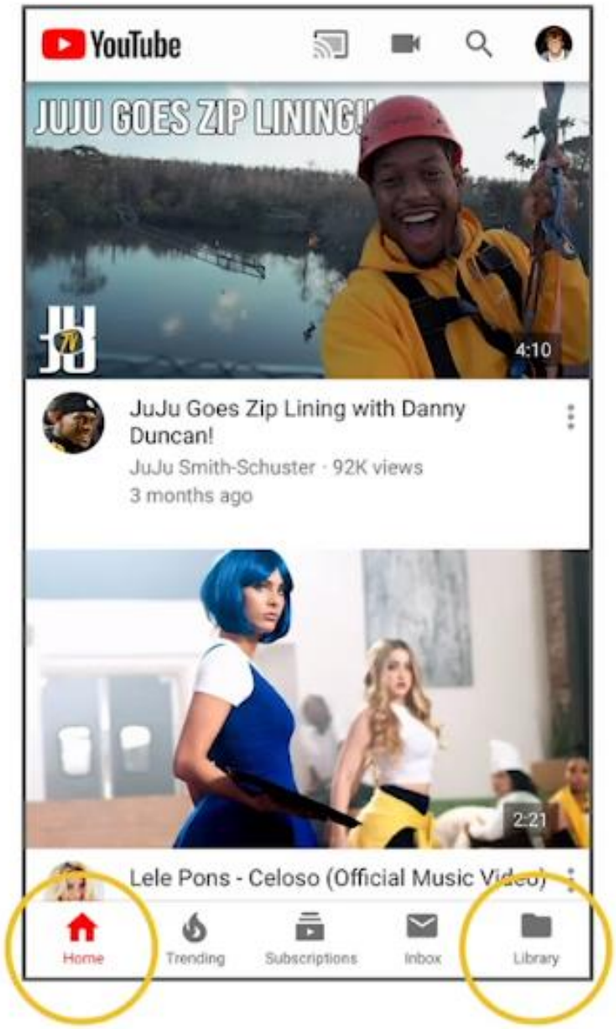


# Psychological phenomena

**2.Serial Position Effect** - This effect describes how people tend to recall the first and last items in a series better than the middle items. This is leveraged in UX design to position crucial actions at strategic places in navigation bars to ensure they are noticed.

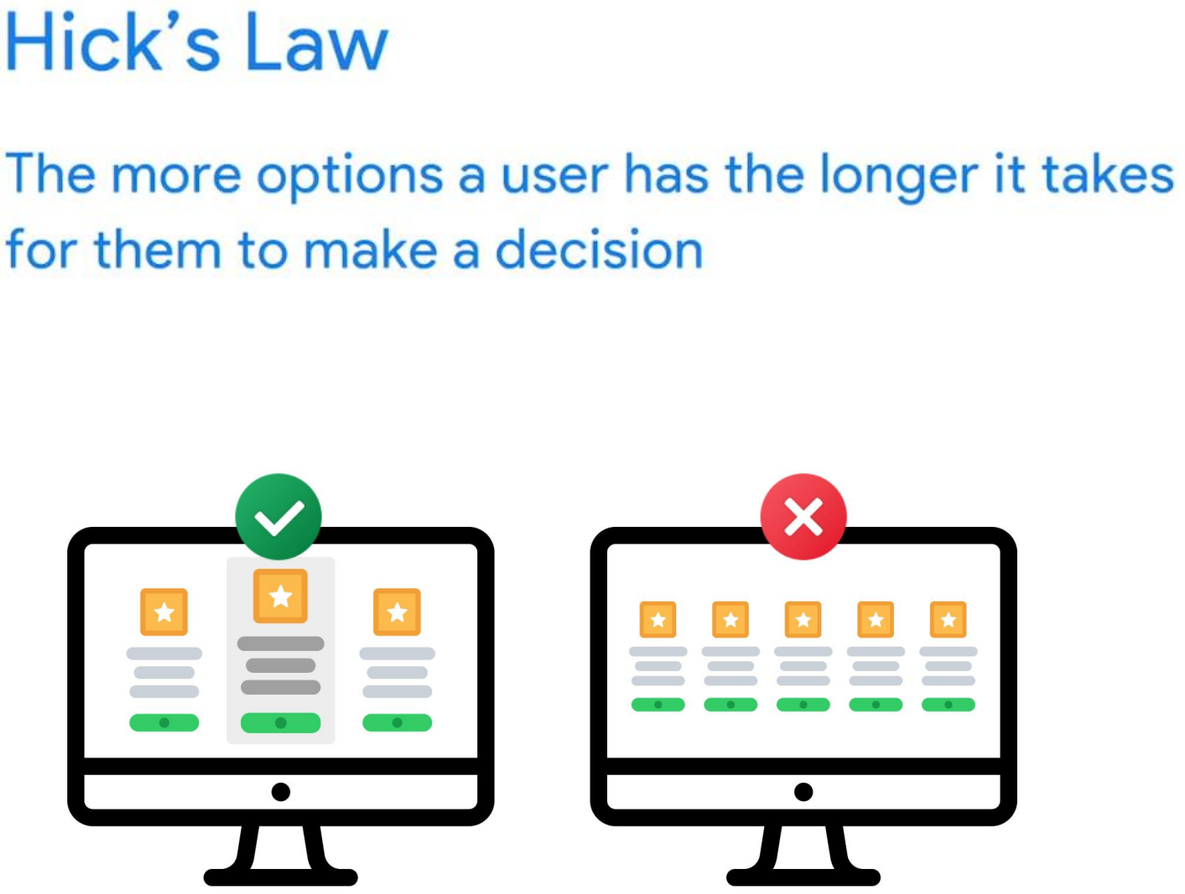


# Psychological phenomena

**3.Hick's Law** - This law states that the more choices presented to a user, the longer it will take for them to make a decision. This is evident in scenarios like choosing among various potato chip flavors, which can overwhelm and prolong the decision-making process.

These psychological insights are crucial for creating effective and user-friendly designs that consider human tendencies and limitations. The challenge for UX designers is to use these principles ethically to enhance user experience without manipulating or overwhelming the user, turning potential design limitations into advantages.



## 2. What is a problem statement?

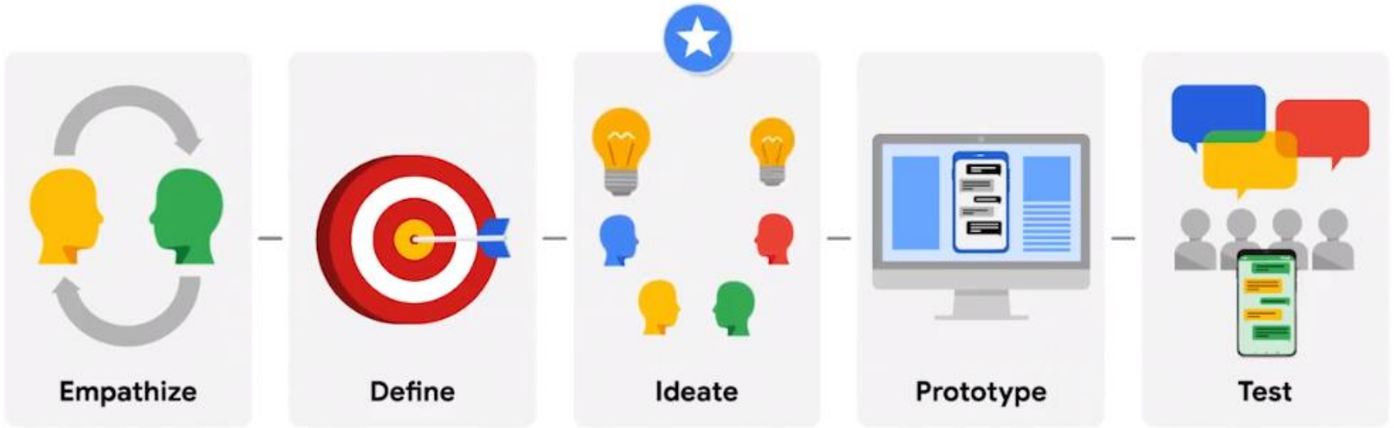
- A problem statement identifies the gap between the current state (i.e. the problem) and the desired state (i.e. the goal) of a process or product. Within the design context, you can think of the user problem as an unmet need. By designing a solution that meets this need, you can satisfy the user and ensure a pleasant user experience.
- A problem statement, or point of view (POV) statement, frames this problem (or need) in a way that is actionable for designers. It provides a clear description of the issue that the designer seeks to address, keeping the focus on the user at all times.

# Understand design ideation

- Ideation is defined as the creative process of generating, developing, and communicating new ideas, where the aim is to produce a large quantity of ideas, exploring all possibilities without immediate critique.
- The discussion highlights various brainstorming techniques such as "How Might We" and "Crazy Eights," encouraging the use of competitive audits and additional research if necessary. A key aspect of ideation is the acceptance that not all ideas generated will be viable, but the practice of generating numerous ideas can lead to innovative solutions. This phase is crucial for transitioning from understanding user needs and problems to actively creating solutions to address those needs.

## Ideation

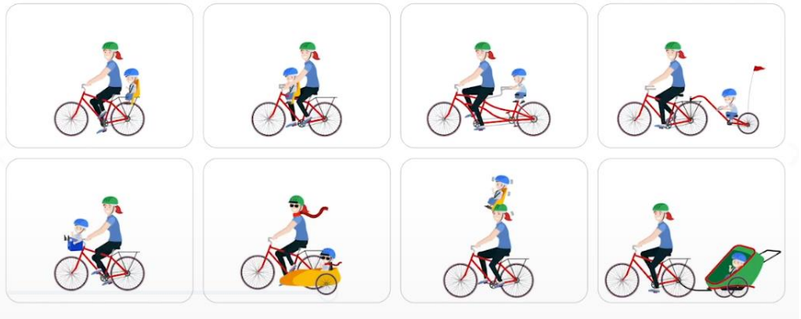
The process of generating a broad set of ideas on a given topic, with no attempt to judge or evaluate them



# ideation

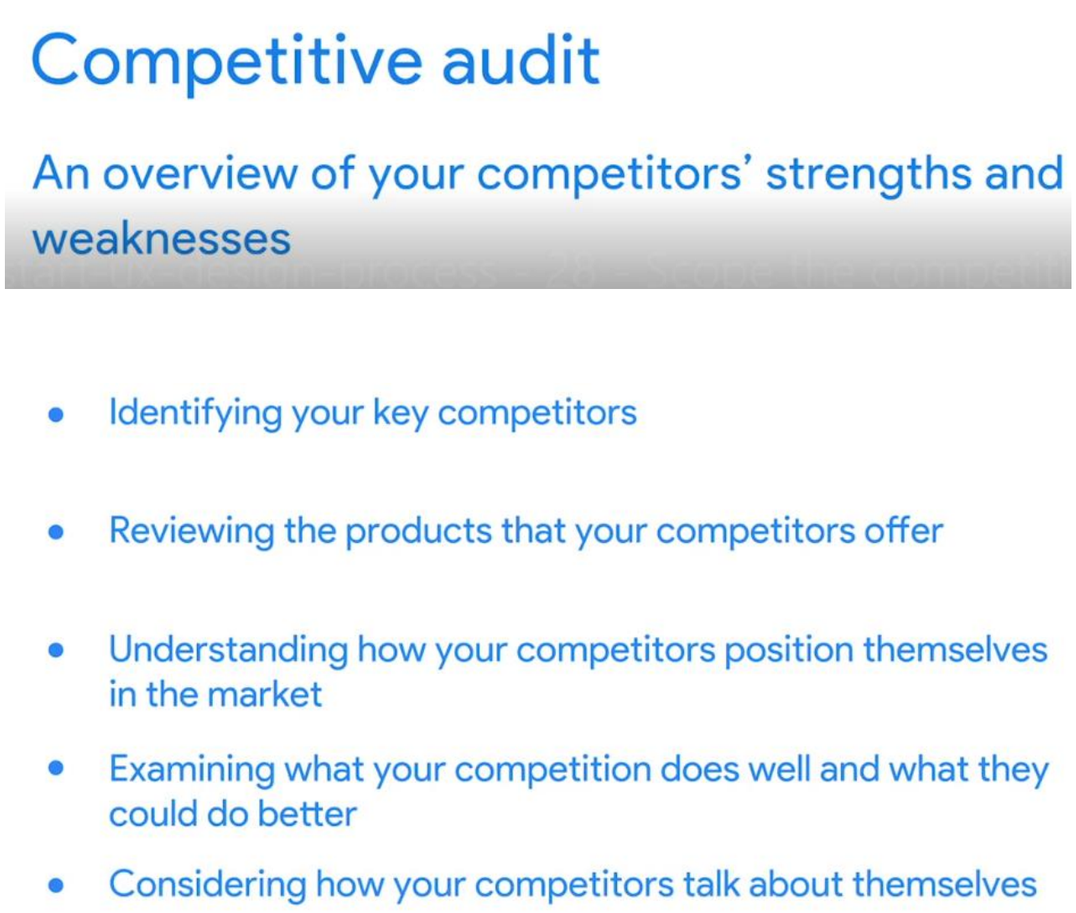
- Ideation is emphasized as a phase where no idea is too outlandish, encouraging a voluminous flow of concepts without immediate judgment.
- This approach ensures that even after constraints like budget or time are applied, there are still ample feasible solutions. Moreover, ideation aims for inclusivity, ensuring the final design is equitable and satisfies the broadest user base possible.
- Finally, it's highlighted that the ultimate choice of solution may not rest solely with the designers but should involve user feedback to determine the most effective design. The goal is to transition from a wide array of ideas to the most practical and user-centered solutions, informed by real-world constraints and user needs.



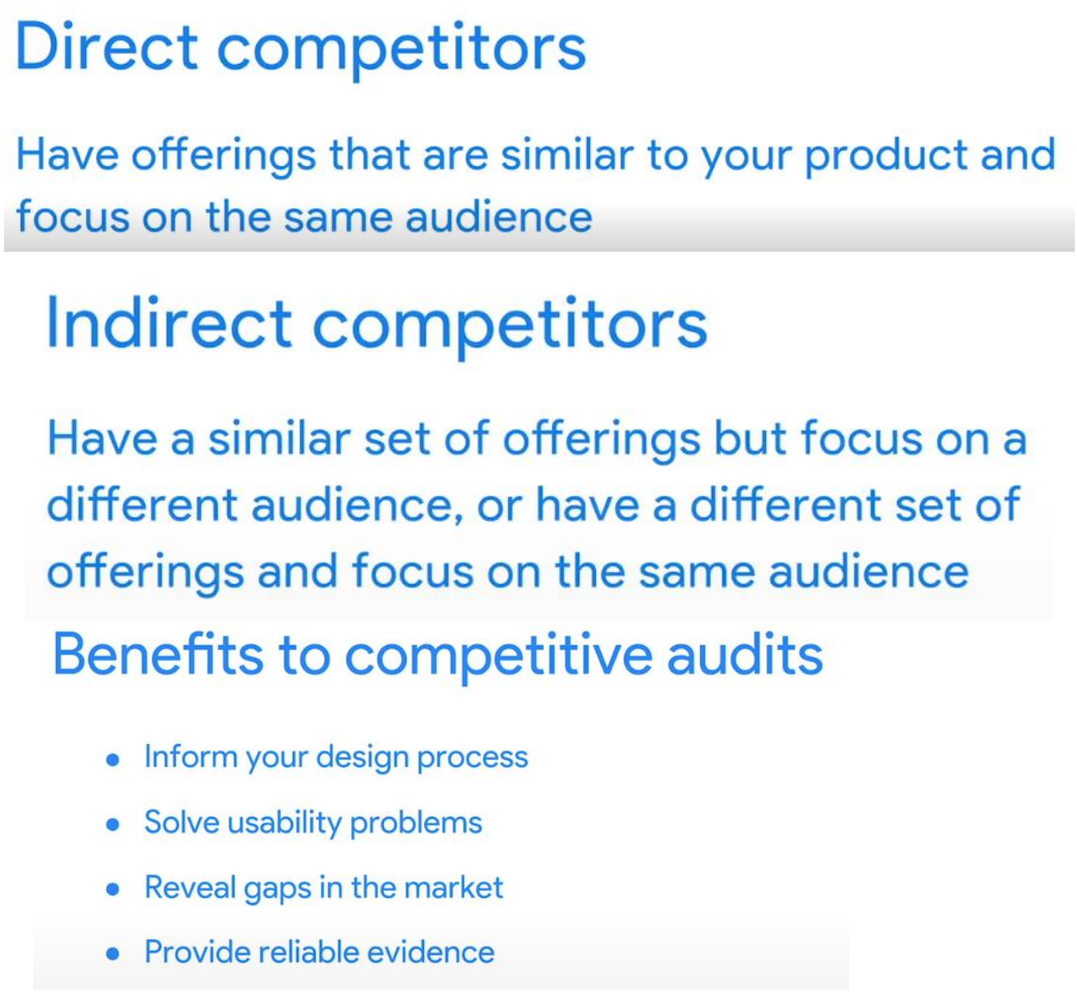
# Scope the competition

- Competitive audits. A competitive audit is an overview of your competitors' strengths and weaknesses.
- This includes: Identifying your key competitors, reviewing the products that your competitors offer, understanding how your competitors position themselves in the market, examining what your competition does well and what they could do better, and considering how your competitors talk about themselves.
- both direct and indirect competitors in your audit



# Scope the competition

- Direct competitors have offerings, meaning products, services, or features, that are similar to your product and focus on the same audience. Essentially, you're both trying to solve the same problem.
- Indirect competitors are a little more nuanced. They either have a similar set of offerings, but focus on a different audience than you, or they have a different set of offerings and focus on the same audience as you. For example, let's say you're creating a weight loss app that focuses on people in their 30s. Your direct competitors are other companies that also make weight loss apps targeted at people in their 30s. Your indirect competitors are any company that makes health or wellness apps targeted at people in their 30s, or make weight loss apps targeted at people in a different age bracket. It's helpful to audit a wide range of products that compete with yours to get a full picture of the landscape.



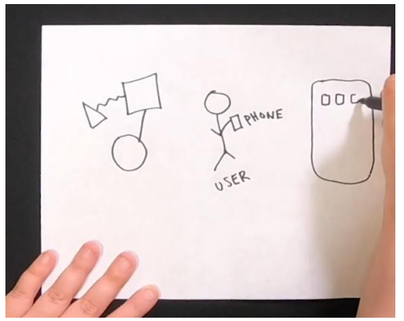
# Scope the competition

## Competitive audit steps

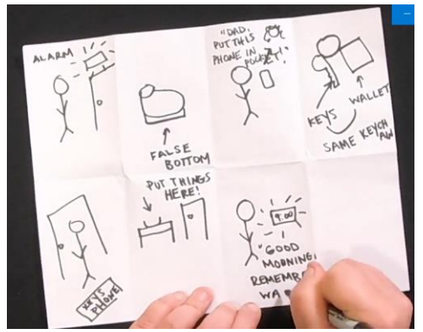
1. Outline the goals
2. Create a spreadsheet with a list of your competitors
3. Call out the specific features you want to compare
4. Research each company
5. Analyze findings

# Method: Use crazy eight for idea

- **Purpose of Crazy Eights**: Allows comparison of ideas, helps see varied concepts, and narrows down options to the best solutions.
- **User-Centric Solutions**: The most effective solution is determined by user preference, not by the designer or team's assumptions.
- **Setup for Crazy Eights**:
  - **Materials Needed**: A large sheet of paper, ideally bigger than regular printer paper. Fold it into eight equal rectangles.
  - **Drawing Tools**: Use Sharpies for clear lines or pencils for adjustable sketches.
  - **Timing**: Allocate eight minutes total, one minute per sketch.
  - **Problem Statement**: Have a clear problem statement to guide the sketches.
- **Example Activity**: Charles needs to keep essential items like his wallet because he tends to lose them. Sketch solutions such as alarms on doors, reminder signs, checklist doormats, and wallet storage in shoes.
- **Execution**: Participants sketch eight solutions in eight minutes, focusing on creativity without restrictions.
- **Feedback and Sharing**: Encouraged to share sketches for feedback and further refinement.





# 4. What is a prototype?

- **Put simply, a prototype is a scaled-down version of your product; a simulation or sample version which enables you to test your ideas and designs *before* investing time and money into actually developing the product. **
- So, if you're designing an app, you might create a digital prototype and test it on real users before **handing it off to the developers**.

## Different kinds of prototypes

- Now let's take a look at some of the different kinds of prototypes you might use. Prototypes can vary in terms of their form, fidelity, interactivity, and lifecycle:
- **Form:** Is it a hand-drawn prototype, or a digital one? Is it for mobile or desktop?
- **Fidelity:** How detailed and polished is the prototype? You'll often hear the terms high-fidelity and low-fidelity in relation to prototypes.

- **Interactivity:** How functional is the prototype? Can the user click on it or interact with it, or is it view-only?
- **Lifecycle:** Is the prototype a quick, disposable version that will be replaced with a new and improved version? Or is it a more enduring creation that can be built and improved upon, potentially ending up as the final product?

So why conduct user testing?

- **User testing saves time and money:** By catching errors and usability issues early on, you ensure that the product you eventually launch is the most bug-free, user-friendly product it can be. What happens if you skip the testing phase in favor of getting the product developed as soon as possible? You'll spend considerable time and money correcting the product post-launch. Not only is this frustrating—it's also bad for the bottom line.
- **User testing reveals unexpected insights:** No matter how thorough your initial **user research** was, or how convinced you are that you've designed the optimal solution to your user's problem, there are always new insights to be uncovered. **Effective usability testing questions** will highlight issues that you would not have noticed otherwise.

- **User testing improves user satisfaction:** Design Thinking is all about putting the user first. By gathering first-hand user feedback, you can make informed design decisions—improving user satisfaction in the long run. As a designer, prototyping and testing will keep you focused on the user at all times. Of course, satisfied customers are good for business, too!

## Remote vs. in-person user testing

- During in-person testing, you'll be in the same room as the user while they test your prototype. This has several advantages. Not only are you able to control the testing environment and keep distractions to a minimum; you can also directly observe the user. You are privy to facial expressions, body language, and any verbal commentary the user makes as they interact with the product—giving you valuable, first-hand insight into their experience. However, in-person testing can be expensive and time-consuming.

# Remote vs. in-person user testing

- Remote user testing offers a less expensive, more convenient alternative, but you'll have little to no control over the user's testing environment. However if you're one of the growing numbers now **working as remote UX designers**, this kind of user testing makes a lot of sense. If you're testing a digital prototype, you can conduct moderated or unmoderated remote user tests. Let's explore each of these options now.

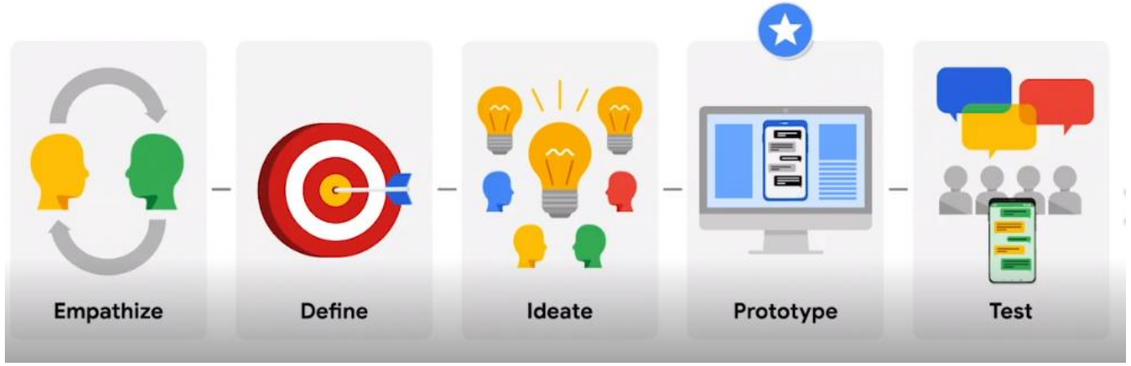
# Moderated vs. unmoderated user testing

- Moderated remote user testing is a good middle ground between in-person tests and completely unmoderated remote tests. Live remote testing allows you to observe your users over a video call, for example. You can use a screen recording app to capture the test, and certain programs will also track and highlight where the user clicks in your digital prototype.

# Build_Wireframes_and_Low-Fidelity_Prototypes

# Prototypes

- use research to inform the ideation process, build storyboards and explain how storyboards are used to create wireframes, draw basic elements like shapes and lines, understand information architecture, create paper wireframes, build digital wireframes on a tool called Figma, construct low-fidelity paper prototypes, develop low-fidelity digital prototypes on Figma, and understand how to avoid the effects of deceptive patterns in designs.
- Keep in mind that throughout this course, you'll continue to build on a project for your portfolio.



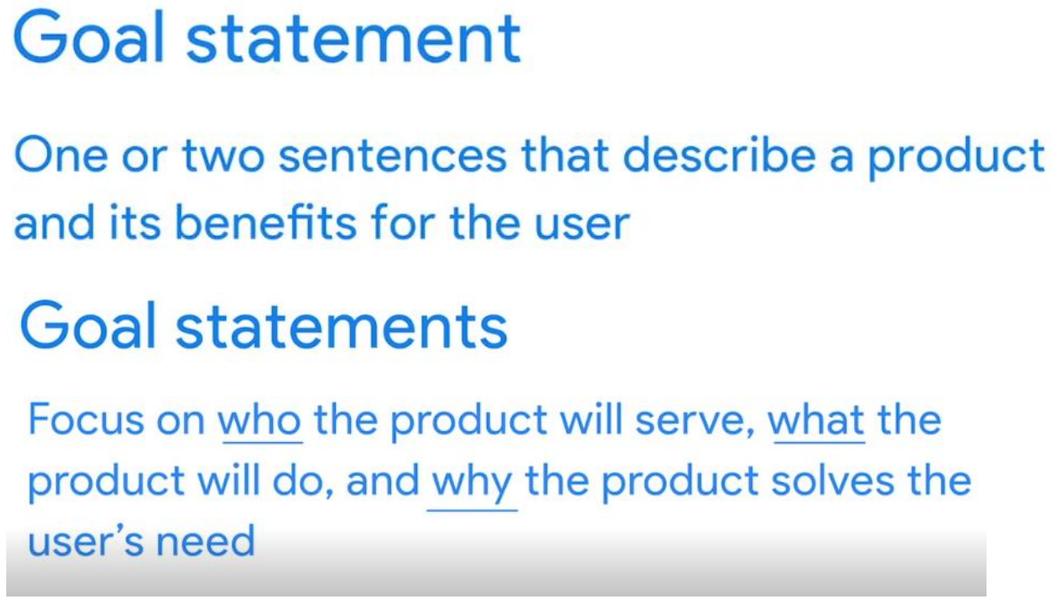

- Use research to inform the ideation process
- Build storyboards and explain how storyboards are used to create wireframes
- Draw basic elements
- Understand information architecture
- Create paper wireframes
- Build digital wireframes on a tool called Figma
- Construct low-fidelity paper prototypes
- Develop low-fidelity digital prototypes on Figma
- Understand how to avoid the effects of deceptive patterns in design

# User Research to inform idea

1. **Empathize with Users:** We began by using empathy maps, personas, user stories, and user journey maps to identify and understand the pain points of users.
2. **Use Tools for Understanding:** We utilized empathy maps to grasp user behaviors and motivations, which include what users say, think, do, and feel, to understand their experiences deeply.
3. **Create Personas:** Personas, fictional characters representing user groups, were developed to focus our design efforts on meeting their specific goals and needs.
4. **Develop User Stories:** User stories, brief narratives from the persona's perspective, were crafted to inspire and guide our design decisions, focusing on essential user needs.
5. **Map User Journeys:** We constructed user journey maps to visualize the sequence of interactions a user has with the product, aiming to address and alleviate pain points effectively.
6. **Synthesize Insights:** All these tools helped us formulate a problem statement that clearly outlines the user needs our design should meet, setting the stage for ideation and solution development.

# Create Goal Statement

- **Creation of a Goal Statement:** This section explains the process of formulating a goal statement. This statement is composed of one or two sentences that articulate the product's utility and its advantages for the user, effectively offering an ideal solution for the design challenges addressed.
- **Defining the Goal Statement:** Here, the conversation focuses on using the problem statement to define who the product is for, what it aims to accomplish, and why it is a valuable solution for the user's needs.
- **Practical Application Example:** We apply the goal statement development process to a hypothetical scenario involving the design of an app for finding and scheduling dog walkers.
- **Linking Insights to Goal Statement:** This part elaborates on how to extract relevant details from the problem statement to clearly define who the product serves, what functionalities it offers, and why these are beneficial to the user.



# Goal Statement

**PROBLEM STATEMENT**

_____ is a _____
user name — user characteristics

who needs _____
user need

because _____ .
insight

**PROBLEM STATEMENT**

Drew is a pet owner in a small town
user name — user characteristics

who needs to find and schedule a dog walker
user need

because they work the night shift .
insight

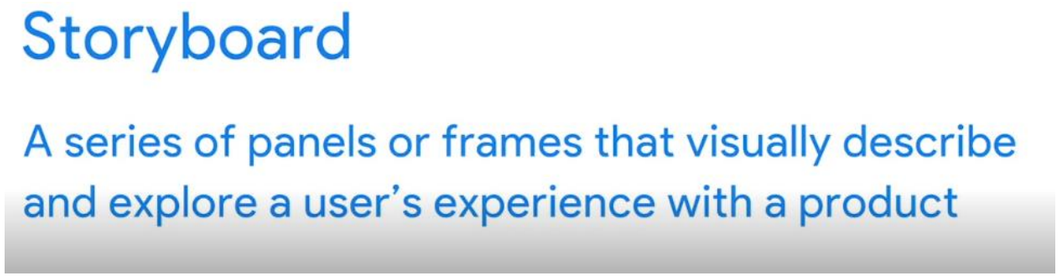
**GOAL STATEMENT**

Help users with pets find and schedule dog walkers quickly and easily

# Storyboard

•A storyboard is a series of panels or frames that visually describe and explore a user's experience with a product. Keep in mind, we've made the transition from the problem the user is facing to coming up with ideas for solutions we can provide as UX designers.

•The four key elements of a storyboard are the character, the scene, the plot, and the narrative. The first element, character, is the user in your story. The next element, the scene, helps us imagine the user's environment. The plot describes the benefit or solution of the design. Finally, the narrative describes the user's need or problem and how the design will solve the problem.

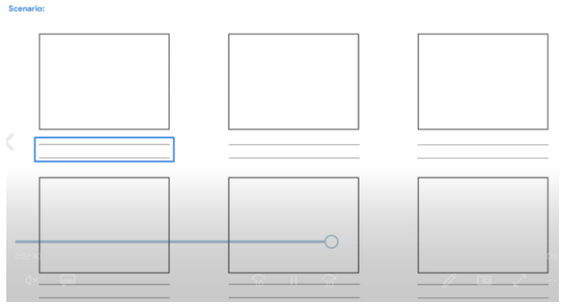


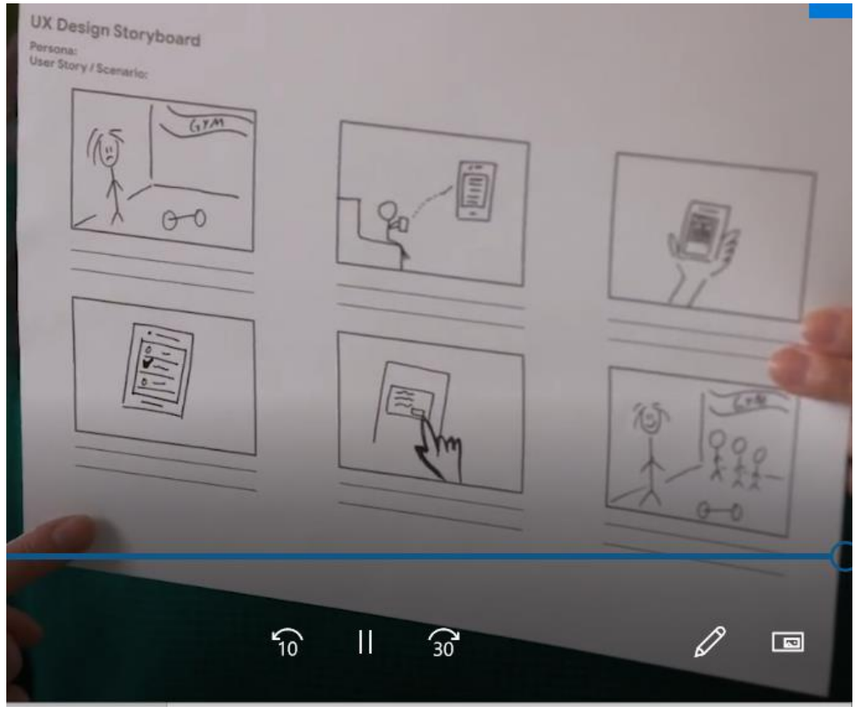



# Storyboard

•Let's revisit a problem statement from course 2. Amal is an athlete who needs a way to sign up for workout classes because the class he wants to participate in fills up fast.

• Our first step is to turn this problem statement into a goal statement. Our goal statement could be: Help users who are athletes sign up for workout classes early and quickly. Now it's time to storyboard.



# Storyboard

- storyboards are often used to outline movies. In the film world, storyboards are used before a movie is created to show how each scene of the movie will play out for a character.
- There are actually two types of storyboards. Keeping movies in mind, these two types of storyboards are called big picture and close-up.

Big picture storyboard

Focuses on the user experience

Close-up storyboard

Focuses on the product

# big-picture storyboard

- First, let's discuss a big-picture storyboard, which focuses on the user experience. Big picture storyboards think about how people will use the product throughout their day and why that product will be useful.
- This helps you understand the entire user experience, including the different challenges, potential pain points, and types of interactions the user will encounter.
- Like a good movie, a big picture storyboard can show the emotional engagement that a user will have with our app or with any product. Understanding how a user feels while experiencing your product is an essential part of the design process.



# close-up storyboard

- The close-up storyboard is less about emotion, since we're not focused on the user. Instead, the close-up storyboard is focused on the practicalities of the design itself. So how do you decide which type of storyboard to use?
- To decide, it's helpful to think about this stage of the design process you're in. If you're early in the design process, you might want to present your high-level ideas to stakeholders to get them excited and bought in.

